A positive-density improvement over the odd numbers in Erdős Problem 327

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Abstract

Let $f_1(N)$ be the largest size of a set $A \subseteq \{1, \dots, N\}$ for which $a + b \nmid ab$ whenever $a, b \in A$ are distinct. The odd numbers show that $f_1(N) \geq N/2 + O(1)$. We prove that there is an absolute $\varepsilon > 0$ such that

$$f_1(N) \geq \left(\frac{1}{2} + \varepsilon\right) N$$

for every sufficiently large N, giving a positive answer to the first question of Erdős Problem 327.

The construction begins with a positive-density set for the stronger condition $a + b \nmid 2ab$, following Sawin's orientation by the total number of prime factors. The new ingredient is to center the prime-factor anatomy at a fixed roughness cutoff and to impose distinct centered budgets at both ends of every odd-even conflict. A three-linear-form mean-value estimate then makes the mixed edge count smaller than the rough source by a power of $\log L$. All cutoff-dependent Euler factors and both residual ranges are computed explicitly. The strict numerical parameter window is certified by rational interval arithmetic. An accompanying Lean 4 development gives an unconditional formal proof of the first conclusion and, separately, of Sawin's positive-density conclusion for $a + b \nmid 2ab$.

1 Introduction

For $k \geq 1$, call a finite set A k-admissible if

$$a + b \nmid kab \qquad (a, b \in A,\ a \neq b).$$

Write

$$f_k(N) = \max\{|A| : A \subseteq \{1, \dots, N\},\ A \text{ is } k\text{-admissible}\}.$$

Erdős and Graham asked whether a 1-admissible set can be "substantially more" than the odd numbers, and whether every 2-admissible set has size $o(N)$ [3, p. 37]. In unit-fraction language, $a + b \mid ab$ precisely when $1/a + 1/b$ is a unit fraction.

Sawin recently answered the second question negatively by constructing 2-admissible sets of positive density [4]. His paper also records a lower bound for the first question, but that bound is below the density 1/2 supplied by the odd numbers. The first question was still listed as open, with no proof claim in its discussion, when the present manuscript was prepared [1].

Our result is the following.

Theorem 1.1. *There are absolute constants $\varepsilon > 0$ and N_0 such that*

$$f_1(N) \geq \left(\frac{1}{2} + \varepsilon\right) N$$

for every $N \geq N_0$.

The source construction also gives an independent proof of Sawin's theorem: there is an absolute $c > 0$ such that $f_2(N) \geq cN$ for every sufficiently large N; see 4.3. Thus the paper and its formalization cover both questions in Problem 327, while the new mathematical conclusion is 1.1.

It is enough to prove a slightly more convenient even-endpoint form.

Theorem 1.2. *There is an absolute $\eta > 0$ such that, for every sufficiently large N, there is a 1-admissible $A_N \subseteq \{1, \ldots, 2N\}$ with*

$$|A_N| \geq (1 + \eta)N.$$

Indeed, apply 1.2 with $N = \lfloor X/2 \rfloor$. For large X, the resulting set lies in $\{1, \ldots, X\}$ and has size at least $(1/2 + \eta/4)X$.

The construction has two layers. First we build a set $B_N \subseteq [N/2, N]$ of size $\gg N/\log L$ such that

$$b + d \nmid 2bd \qquad (b, d \in B_N,\ b \neq d).$$

Thus $2B_N$ has no even-even conflict for the original divisibility condition. Second, we begin with almost every odd integer $a \leq 2N$ and delete those incident to a mixed conflict

$$a + 2b \mid ab, \qquad b \in B_N.$$

The main work is to make both the deleted odd set and the mixed edge set $o_L(N/\log L)$, while $B_N \gg N/\log L$.

The centering

$$\Omega_L(n, x) \leq A \log \frac{\log x}{\log L} + K$$

is essential. An L-rough number has spent no prime-factor budget below L, and the subtraction of $\log \log L$ is exactly what leaves a negative power of $\log L$ after the dyadic sums.

2 Notation and analytic inputs

All logarithms are natural. Let $\Omega(n)$ denote the total number of prime factors of n, counted with multiplicity. For real $L > 2$, put

$$\Omega_L(n, x) = \sum_{\substack{p^\nu \| n \\ L \leq p \leq x}} \nu, \qquad \rho_L = \prod_{p < L} \left(1 - \frac{1}{p}\right).$$

We say that n is L-rough when it has no prime divisor below L; we set $P^-(1) = +\infty$.

We use two mean-value results. The first is the nonnegative multiplicative-function estimate quoted as Theorem III.3.5 by Tenenbaum [5] and used in Sawin's Lemmas 4 and 5.

Proposition 2.1 (one-variable mean value)**.** *Let g be a nonnegative multiplicative function. Suppose that, for all $y \geq 1$,*

$$\sum_{p \leq y} g(p) \log p \leq C_1 y$$

and

$$\sum_p \sum_{\nu \geq 2} \frac{g(p^\nu) \log(p^\nu)}{p^\nu} \leq C_2.$$

Then

$$\sum_{n\le Y} g(n) \ll (C_1+C_2+1)Y \prod_{p\le Y}\left(1-\frac{1}{p}+\sum_{\nu\ge 1}\frac{g(p^\nu)}{p^\nu}\right),$$

with an absolute implied constant.

The second input is Theorem 3.1 of de la Bretèche and Tenenbaum [2]. We only need its specialization to three fixed primitive linear forms in two variables. We record the features needed to make the cutoff uniformity transparent.

Let $Q_1, Q_2, Q_3 \in \mathbb{Z}[U,V]$ be fixed, primitive, pairwise nonproportional linear forms. For a nonnegative multiplicative function $F : \mathbb{N}^3 \to \mathbb{R}_{\ge 0}$, write $F \in \mathcal{M}_3(1,1,\epsilon)$ if, whenever $(a_1a_2a_3, b_1b_2b_3) = 1$,

$$F(a_1b_1, a_2b_2, a_3b_3) \le F(b_1, b_2, b_3).$$

This is the special case of their class that we use.

For a prime p, let

$$\rho_Q^+(p) = \#\{(u,v) \bmod p : p \mid Q_1(u,v)Q_2(u,v)Q_3(u,v)\}.$$

The theorem supplies, on every box of side lengths comparable to X, an upper bound of the form

$$\sum_{(u,v)\text{ in the box}} F(Q_1(u,v), Q_2(u,v), Q_3(u,v)) \ll X^2 E_{F,Q}(CX) \prod_{3<p\le CX}\left(1-\frac{\rho_Q^+(p)}{p^2}\right), \tag{1}$$

where C depends only on the fixed box aspect ratio and the forms. The factor $E_{F,Q}$ is a nonnegative truncated multiplicative sum. Consequently it is at most the product of its local prime-power factors. For the triples used below, a local pattern with exactly one positive valuation e has density

$$\frac{(1-p^{-1})^2}{p^e} \tag{2}$$

at every odd prime, and patterns with exactly two positive valuations vanish. We calculate every remaining local weight explicitly in the two applications.

The constant in (1) is uniform over $F \in \mathcal{M}_3(1,1,\epsilon)$ once the forms, class parameters, and box aspect ratio are fixed. This is the uniformity that permits F to depend on the cutoff L.

3 Centered prime-factor anatomy

Call n (L,A,K)-regular if

$$\Omega_L(n,x) \le A\log\frac{\log x}{\log L} + K \qquad (x \ge L). \tag{3}$$

Lemma 3.1 (centered maximal tail)**.** *Fix $1 < z < 2$ and $A > 0$ with*

$$A\log z > z - 1.$$

Uniformly for real $L > 2$, $Y \ge 1$, and $K \ge 0$,

$$\#\{n \le Y : n \textit{ is not } (L,A,K)\textit{-regular}\} \ll_{A,z} Yz^{-K}.$$

Among L-rough integers, the stronger bound

$$\#\{n \le Y : P^-(n) \ge L,\ n \textit{ is not regular}\} \ll_{A,z} Y\rho_L z^{-K}$$

holds with the same uniformity.

Proof. Fix $x \geq L$ and put $X = \min(x, Y)$. Apply 2.1 to

$$g(n) = z^{\Omega_L(n,x)}.$$

At $L \leq p \leq X$, its Euler factor is

$$1 - \frac{1}{p} + \frac{z}{p - z} = 1 + \frac{z-1}{p} + O_z(p^{-2}),$$

while outside this range it is $1 + O(p^{-2})$. Because every weighted prime is at least $3 > z$, the prime-power tail is bounded uniformly in L, x; the prime hypothesis follows from Chebyshev's estimate. Mertens' estimate gives

$$\sum_{n \leq Y} z^{\Omega_L(n,x)} \ll_z Y \left(\frac{\log X}{\log L} \right)^{z-1} \tag{4}$$

when $L \leq Y$, with the ratio interpreted as 1 at the initial endpoint.

For

$$g(n) = \mathbf{1}_{P^-(n) \geq L} z^{\Omega_L(n,x)},$$

the factor at $p < L$ is exactly $1 - p^{-1}$. Hence

$$\sum_{\substack{n \leq Y \\ P^-(n) \geq L}} z^{\Omega_L(n,x)} \ll_z Y \rho_L \left(\frac{\log X}{\log L} \right)^{z-1}. \tag{5}$$

If $L > Y$, the exceptional set is empty, so no separate range remains.

Choose $\theta > 0$ so small that

$$\frac{A \log z}{1 + \theta} > z - 1,$$

and put

$$r_j = (1 + \theta)^j, \qquad x_j = L^{\exp(r_j)}.$$

If (3) fails for some $x \in [L, x_0]$, then $\Omega_L(n, x_0) > K$. If it fails for $x \in [x_{j-1}, x_j]$, $j \geq 1$, monotonicity gives

$$\Omega_L(n, x_j) > \frac{A}{1 + \theta} r_j + K.$$

Markov's inequality and (4) bound the latter exceptional set by

$$\ll_z Y z^{-K} \exp\left\{ \left(z - 1 - \frac{A \log z}{1 + \theta} \right) r_j \right\}.$$

The sum over j converges. For $n \leq Y$, the left side of (3) is constant once $x \geq Y$, while the threshold increases, so the grid may be stopped after crossing Y. Replacing (4) by (5) proves the rough version. $\square$

Remark 3.2. The restriction on z is substantive. For example, at $L = 3$ a bound uniform in K cannot have decay z^{-K} when $z > 3$, because multiples of 3^{K+1} already violate the initial threshold. The range $1 < z < 2$ is more than sufficient below.

4 A centered two-admissible source

Fix $\delta \in (0,1)$, $1 < A < 3/(2\log 4)$, and $K \geq 0$. Let

$$S = S(N; L, A, K) = \{b \in [\delta N, N] : P^-(b) \geq L,\ b \text{ is } (L, A, K)\text{-regular}\}.$$

Lemma 4.1 (source density)**.** *There is $K_0 = K_0(A, \delta)$ such that, if $K \geq K_0$, then for every fixed $L > 2$ and all sufficiently large N,*

$$|S| \geq \frac{1-\delta}{2} N\rho_L.$$

The choice of K_0 is independent of L.

Proof. For fixed L, inclusion-exclusion gives

$$\#\{b \in [\delta N, N] : P^-(b) \geq L\} = (1-\delta)N\rho_L + O_L(1).$$

Choose $z \in (1,2)$ close enough to 1 that $A\log z > z - 1$. By 3.1, the excluded regularity tail is at most $C_{A,z}N\rho_L z^{-K}$. Choose K_0 so that $C_{A,z}z^{-K_0} \leq (1-\delta)/4$, and then let N absorb the $O_L(1)$ term. □

For $b \in S$, call b bad if there is $c \leq N$, $c \neq b$, with

$$b + c \mid 2bc, \qquad \Omega(c) \leq \Omega(b). \tag{6}$$

Let $T(N; L, A, K)$ be the number of bad b.

Proposition 4.2 (centered source estimate)**.** *Put $M = A\log 4$. Then*

$$\frac{T(N; L, A, K)}{N\rho_L} \ll_{A,\delta} 4^K \left\{(\log L)^{-3/2} + (\log L)^{1/4-M}(\log N)^{M-3/2}\right\}. \tag{7}$$

The estimate holds for every fixed K, L and all sufficiently large N. If $K \geq K_0(A,\delta)$, division by 4.1 gives the same right side, up to a constant, as an upper bound for the proportion of S that is bad.

Proof. Let b, c satisfy (6), put

$$u = \frac{b}{(b,c)}, \qquad v = \frac{c}{(b,c)}.$$

Then

$$(u,v) = 1, \quad u(u+v) \mid 2b, \quad v \leq \delta^{-1}u, \quad \Omega(v) \leq \Omega(u), \quad (u,v) \neq (1,1). \tag{8}$$

Indeed, $u + v \mid 2(b,c)uv$, while $(u+v, uv) = 1$; multiplying by u gives the second condition. The remaining statements follow from $c/b = v/u$, the range of b, c, and complete additivity of Ω. These are Sawin's coordinates [4, Lemma 2].

Write

$$2b = u(u+v)d, \qquad x = u + v.$$

Because b is L-rough and $L > 2$, b is odd. No odd prime below L divides $u(u+v)d$, and $v_2(u(u+v)d) = 1$. Every nonzero summand has $x \geq L$: the case $x = 2$ is excluded, while $2 < x < L$ would introduce either an odd prime below L or a factor 4.

With $e = v_2(x) \in \{0,1\}$, the exact identity is

$$\Omega_L(b,x) = \Omega(u) + \Omega(x) - e + \Omega_L(d,x).$$

Thus regularity at x gives

$$\Omega(u)+\Omega(x)+\Omega_L(d,x)\le A\log\frac{\log x}{\log L}+K+1.$$

Exponentiating with base 4 bounds the indicator by

$$4^{K+1}\left(\frac{\log x}{\log L}\right)^M 4^{-\Omega(u)}4^{-\Omega(x)}4^{-\Omega_L(d,x)}. \tag{9}$$

We next sum d. For $D\ge 1$, let

$$R(D;x,L)=\sum_{\substack{d\le D\\ p\nmid d\ (2<p<L)}} 4^{-\Omega_L(d,x)}.$$

Apply 2.1. The local factors are

$$1-\frac{1}{p}\quad (2<p<L),$$

$$1-\frac{1}{p}+\frac{1}{4p-1}=1-\frac{3}{4p}+O(p^{-2})\quad (L\le p\le x),$$

and $1+O(p^{-2})$ elsewhere; $p=2$ contributes a fixed factor. Splitting according to $D\ge x$ or $D<x$ yields

$$R(D;x,L)\ll D\left\{(\log x)^{-3/4}(\log L)^{-1/4}+(\log(3D))^{-3/4}\right\}. \tag{10}$$

It remains to average the u,v,x weights. Define

$$g_{1,L}(n)=2^{-\Omega(n)}\mathbf{1}_{p\nmid n\ (2<p<L)}\mathbf{1}_{4\nmid n},$$

$$g_2(n)=2^{-\Omega(n)},$$

and

$$g_{3,L}(n)=4^{-\Omega(n)}\mathbf{1}_{p\nmid n\ (2<p<L)}\mathbf{1}_{4\nmid n}.$$

Since $\Omega(v)\le\Omega(u)$,

$$4^{-\Omega(u)}4^{-\Omega(u+v)}\le g_{1,L}(u)g_2(v)g_{3,L}(u+v).$$

Encode $(u,v)=1$ by

$$F_L(n_1,n_2,n_3)=\mathbf{1}_{(n_1,n_2)=1}g_{1,L}(n_1)g_2(n_2)g_{3,L}(n_3).$$

For coprime tuples $\mathbf{a},\mathbf{b}$,

$$F_L(\mathbf{ab})=F_L(\mathbf{a})F_L(\mathbf{b})\le F_L(\mathbf{b}),$$

including the cases in which a support condition makes one side zero. Hence $F_L\in\mathcal{M}_3(1,1,\epsilon)$ uniformly in L.

Apply (1) to $U,V,U+V$. At every prime,

$$\rho_Q^+(p)=3p-2.$$

At odd $p\ge L$, (2) gives the exact local factor

$$1+(1-p^{-1})^2\left(\frac{1}{2p-1}+\frac{1}{2p-1}+\frac{1}{4p-1}\right)=1+\frac{5}{4p}+O(p^{-2}).$$

For $2 < p < L$, only the V-axis survives, giving

$$1 + \frac{(1-p^{-1})^2}{2p-1} = 1 + \frac{1}{2p} + O(p^{-2}).$$

At $p = 2$, the U- and $U+V$-valuations are at most one, while the V-valuation is arbitrary. Its exact factor is

$$1 + \frac{1}{16} + \frac{1}{32} + \frac{1}{12} = \frac{113}{96}.$$

All p^{-2} constants are absolute. Consequently the Euler factor in (1) is

$$\ll (\log X)^{5/4} (\log L)^{-3/4},$$

while the root-sieve product is $\ll (\log X)^{-3}$. Therefore, on $u \in [X, 2X]$,

$$\sum_{\substack{X \le u < 2X \\ 0 < v \le \delta^{-1} u \\ (u,v)=1}} g_{1,L}(u) g_2(v) g_{3,L}(u+v) \ll_\delta X^2 (\log X)^{-7/4} (\log L)^{-3/4}. \tag{11}$$

Every nonzero block has $X \ge L$. Indeed $u \mid b$ and b is L-rough, so u is L-rough; and $u = 1$ would force $\Omega(v) \le \Omega(u) = 0$, hence $v = 1$ and $b = c$, which (8) excludes. So $u \ne 1$ and $u \ge L$.

Use (9) and (10) with $D = 2N/[u(u+v)]$, and sum (11) dyadically. The first residual term, after division by $N\rho_L$, contributes on one block

$$\ll_{A,\delta} 4^K (\log L)^{-M} (\log X)^{M-5/2}.$$

Since $M < 3/2$,

$$\sum_{X \text{ dyadic}, \ X \gg L} (\log X)^{M-5/2} \ll (\log L)^{M-3/2},$$

which gives $4^K (\log L)^{-3/2}$.

The second residual term contributes

$$\ll_{A,\delta} 4^K (\log L)^{1/4-M} (\log X)^{M-7/4} \left(\log \frac{3N}{X^2} \right)^{-3/4}.$$

Here $X^2 \le u(u+v) \ll_\delta X^2$, while a nonempty term has $u(u+v) \le 2N$. Thus both endpoint logarithms are bounded below by a positive constant; when large they differ by $O_\delta(1)$, and when bounded their negative powers are $O_\delta(1)$. This justifies the displayed replacement of $u(u+v)$ by X^2. Put $T = \log N$, $t = \log X$, and $h = M - 7/4$. For $t \le T/4$, the last logarithm is comparable to T, so the dyadic sum is

$$\ll T^{-3/4} T^{h+1} = T^{M-3/2}.$$

For $t > T/4$, one has $t \asymp T$, while the endpoint logarithm moves in constant increments and

$$\sum_{1 \le r \ll T} r^{-3/4} \ll T^{1/4}.$$

This again gives $T^{M-3/2}$, proving (7). $\square$

Delete every bad element of S, and call the surviving set B. If distinct $b, d \in B$ satisfied $b + d \mid 2bd$, one of $\Omega(b), \Omega(d)$ would be at least the other, so the endpoint with the larger value would have been deleted. Hence B is 2-admissible.

Corollary 4.3 (Sawin's conclusion)**.** *There is an absolute constant $c > 0$ such that*

$$f_2(N) \geq cN$$

for every sufficiently large N.

Proof. Fix $\delta = 1/2$ and $1 < A < 3/(2 \log 4)$, and choose K from 4.1. Because $A \log 4 < 3/2$, the first term in (7) is made smaller than one quarter of the source density by fixing L sufficiently large. With that L fixed, the second term tends to zero as N tends to infinity. The surviving set B is therefore 2-admissible and has size at least $N\rho_L/8$ for all sufficiently large N. Taking $c = \rho_L/8 > 0$ proves the claim. $\square$

5 The two-endpoint mixed estimate

We now estimate the conflicts between an odd host and twice the rough source.

Lemma 5.1 (mixed coordinates)**.** *Let a, b be odd. Then*

$$a + 2b \mid ab$$

if and only if there are unique $t, u, w \in \mathbb{N}$ such that

$$a = tw(2u + w), \qquad b = tu(2u + w), \qquad (u, w) = 1.$$

Proof. Put $g = (a, b)$, $a = gw$, and $b = gu$. Then $(u, w) = 1$, and since w is odd,

$$(2u + w, uw) = 1.$$

Thus $g(2u + w) \mid g^2uw$ if and only if $2u + w \mid g$. Writing $g = t(2u + w)$ gives the representation and its uniqueness. The converse is immediate. $\square$

Fix positive A_b, A_o, K_b, K_o, bases $q_b, q_o > 1$, and $\delta \in (0, 1)$. Let $S \subseteq [\delta N, N]$ consist of L-rough integers satisfying the centered (A_b, K_b) bound. Let O consist of odd $a \leq 2N$ satisfying the centered (A_o, K_o) bound. Define

$$E_{\mathrm{mix}}(N) = \#\{(a, b) \in O \times S : a + 2b \mid ab\}.$$

Put

$$\alpha = q_b^{-1}, \qquad \beta = q_o^{-1}, \qquad s = (q_b q_o)^{-1},$$

$$M = A_b \log q_b + A_o \log q_o,$$

and

$$E = M + \alpha + \beta + 2s - 4.$$

Proposition 5.2 (two-endpoint estimate)**.** *If $E < -1$, then*

$$E_{\mathrm{mix}}(N) \ll q_b^{K_b} q_o^{K_o} \left\{ N(\log L)^{-2} + o_L(N) \right\}. \tag{12}$$

The normalized o_L-function can be chosen independently of K_b, K_o.

Proof. Use 5.1 and put $x = 2u + w$. Since $b = tux$ is L-rough and a, b are odd, all of t, u, w, x are odd and t, u, x are L-rough. The endpoint ranges imply

$$\frac{w}{u} = \frac{a}{b} \leq \frac{2}{\delta}, \qquad \frac{\delta N}{ux} \leq t \leq \frac{N}{ux}.$$

We first record the lower bound on u that starts the dyadic sum. Assume $L > 2 + 2\delta^{-1}$, as we may, since L is at our disposal. Then

$$u \geq L, \qquad \text{and hence} \qquad x > 2u \geq 2L. \tag{13}$$

Indeed $x \mid b$ and b is L-rough, so x is L-rough; as $x = 2u + w \geq 3$ we get $x \geq L$. If u were 1, the range above would give $w = a/b \leq 2\delta^{-1}$ and therefore $x = 2 + w \leq 2 + 2\delta^{-1} < L$, a contradiction. So $u \neq 1$, and u is L-rough, whence $u \geq L$.

Equivalently, in the language of the multiplier: at $k = 1$ the greatest common divisor $g = tx$ already contains all of x, so $u = 1$ would force x to divide $k = 1$. It is this feature that fails for a general multiplier, where only $x/(x, k)$ need divide g. At the common scale x,

$$\Omega_L(b, x) = \Omega_L(t, x) + \Omega(u) + \Omega(x),$$

$$\Omega_L(a, x) = \Omega_L(t, x) + \Omega_L(w, x) + \Omega(x) \geq \Omega_L(t, x) + \Omega_{\geq L}(w) + \Omega(x) - 1. \tag{14}$$

The inequality in (14) is where a prime factor of w exceeding x is discarded, and one such factor is all that can occur: by (13) we have $x \geq L > 2\delta^{-1}$, while $w \leq 2\delta^{-1}u < 2\delta^{-1}x$, so two prime factors above x would force $w > x^2 > 2\delta^{-1}x$. Exponentiating both regularity bounds therefore majorizes the joint indicator by

$$q_o\, q_b^{K_b} q_o^{K_o} \left(\frac{\log x}{\log L}\right)^M \alpha^{\Omega(u)} \beta^{\Omega_{\geq L}(w)} s^{\Omega(x)} s^{\Omega_L(t,x)}. \tag{15}$$

The single extra factor q_o is an absolute constant: it does not depend on L, N, K_b or K_o, it is absorbed by the implied constant of (12), and it changes no exponent. In particular the parameter window of 6 and the certified margins of 1 are untouched.

For $Y = N/(ux)$, 2.1 applied to

$$h_{L,x}(t) = \mathbf{1}_{P^-(t) \geq L} s^{\Omega_L(t,x)}$$

gives

$$\sum_{\substack{\delta Y \leq t \leq Y \\ P^-(t) \geq L}} s^{\Omega_L(t,x)} \ll_s Y \left\{ (\log x)^{s-1} (\log L)^{-s} + (\log(3Y))^{s-1} \right\}. \tag{16}$$

Indeed, the local factors are $1 - p^{-1}$ below L,

$$1 - \frac{1-s}{p} + O_s(p^{-2})$$

for $L \leq p \leq \min(x, Y)$, and $1 + O(p^{-2})$ elsewhere. If $Y < L$, the only possible L-rough integer up to Y is 1, which is covered by the second term.

For the outer variables define

$$g_{1,L}(n) = \mathbf{1}_{P^-(n) \geq L} \alpha^{\Omega(n)},$$

$$g_{2,L}(n) = \mathbf{1}_{2\nmid n}\beta^{\Omega_{\geq L}(n)},$$

$$g_{3,L}(n) = \mathbf{1}_{P^-(n)\geq L}s^{\Omega(n)},$$

and

$$F_L(n_1, n_2, n_3) = \mathbf{1}_{(n_1,n_2)=1}g_{1,L}(n_1)g_{2,L}(n_2)g_{3,L}(n_3).$$

As in the source estimate, $F_L \in \mathcal{M}_3(1, 1, \epsilon)$ uniformly in L.

Apply (1) to

$$Q_1(U, W) = U, \qquad Q_2(U, W) = W, \qquad Q_3(U, W) = 2U + W.$$

At odd primes the three root lines are distinct and have union size

$$\rho_Q^+(p) = 3p - 2.$$

At $p = 2$, all three coordinate values are forced odd, so the exact local factor is 1. For $2 < p < L$, only the W-axis is permitted, and (2) gives

$$\mathcal{E}_p = 1 + (1 - p^{-1})^2 \sum_{e\geq 1} p^{-e} = 1 + \frac{1}{p} - \frac{1}{p^2}. \tag{17}$$

For $p \geq L$, the complete factor is

$$\mathcal{E}_p = 1 + (1 - p^{-1})^2 \left(\frac{\alpha}{p - \alpha} + \frac{\beta}{p - \beta} + \frac{s}{p - s}\right). \tag{18}$$

Thus

$$\mathcal{E}_p = 1 + \frac{\alpha + \beta + s}{p} + O_{\alpha,\beta,s}(p^{-2})$$

above L, with an error constant independent of L. Mertens' estimate and the root-sieve product yield, on $u \in [X, 2X]$,

$$\sum_{\substack{X\leq u<2X\\ 0<w\leq(2/\delta)u\\ (u,w)=1}} \alpha^{\Omega(u)}\mathbf{1}_{2\nmid w}\beta^{\Omega_{\geq L}(w)}s^{\Omega(2u+w)}\mathbf{1}_{P^-(u),P^-(2u+w)\geq L} \ll X^2(\log X)^{\alpha+\beta+s-3}(\log L)^{1-\alpha-\beta-s}. \tag{19}$$

Every relevant block has $X \geq L$ by (13), and $ux \asymp_\delta X^2$.

Insert the first term of (16) into (15) and use (19). One block contributes

$$\ll q_b^{K_b} q_o^{K_o} N(\log X)^E(\log L)^{-E-3}.$$

Since $E < -1$, the dyadic sum from $X \gg L$ is

$$\ll (\log L)^{E+1},$$

leaving exactly

$$q_b^{K_b} q_o^{K_o} N(\log L)^{-2}.$$

For the second residual term, put

$$H = M + \alpha + \beta + s - 3.$$

One block contributes

$$\ll q_b^{K_b} q_o^{K_o} N(\log L)^{1-\alpha-\beta-s-M}(\log X)^H \left(\log \frac{3N}{X^2}\right)^{s-1}. \tag{20}$$

The replacement of ux by X^2 inside the logarithm is legitimate despite the negative exponent. Namely, if $Z = ux/X^2$, then $1 < Z \le C_\delta$; writing $N/X^2 = ZV$, $V \ge 1$, shows

$$\log \frac{3N}{ux} = \log 3 + \log V \ge c_\delta \log \frac{3N}{X^2}.$$

Now $H + s = E + 1 < 0$. Write $U = \log N$, $y = \log X$, and split at $y = U/4$. Below the split, the endpoint logarithm is comparable to U. If $H > -1$, the normalized sum in (20) is

$$O(U^{s-1}U^{H+1}) = O(U^{E+1}) = o(1).$$

For $H = -1$, only an extra $\log U$ appears, and for $H < -1$ the y-sum is bounded at its fixed L-dependent lower endpoint; both are again $o(1)$. Above the split, $y \asymp U$, and because $s - 1 \in (-1, 0)$, summing the endpoint variable gives $O(U^s)$. The result is again $O(U^{H+s}) = O(U^{E+1}) = o(1)$.

The factor $q_b^{K_b} q_o^{K_o}$ was outside every normalized sum, so the remaining o_L-function is independent of K_b, K_o. This proves (12). □

6 A feasible parameter window

We use the rational decimal parameters

$$\begin{aligned} A_b &= 1.000001, & q_b &= 2.48933, \\ A_o &= 1.16312, & q_o &= 1.34288, \\ z_o &= 1.34305, & c &= 3.3912. \end{aligned} \tag{21}$$

They satisfy

$$\begin{aligned} A_b \log 4 &< \frac{3}{2}, \\ A_o \log z_o &> z_o - 1, \\ c \log z_o &> 1, \\ c \log q_o &< 1, \\ A_b \log q_b + A_o \log q_o + q_b^{-1} + q_o^{-1} + 2(q_b q_o)^{-1} - 4 &< -1. \end{aligned} \tag{22}$$

For reproducibility, the supplement bounds every logarithm by rational intervals obtained from

$$\log x = 2 \sum_{j \ge 0} \frac{y^{2j+1}}{2j+1}, \qquad y = \frac{x-1}{x+1},$$

with an exact positive-tail bound. The certified margins are shown in 1.

7 Proof of the main theorem

Proof of 1.2. Fix the parameters in (21) and set $\delta = 1/2$. Choose $z_b \in (1, 2)$ sufficiently close to 1 that

$$A_b \log z_b > z_b - 1.$$

quantity	certified one-sided margin
$A_b \log 4 - 5/2 < -1$	upper value < -1.1137042525
$A_o \log z_o - (z_o - 1) > 0$	> 0.0000042730
$c \log z_o - 1 > 0$	> 0.0002111998
$1 - c \log q_o > 0$	> 0.0002180771
mixed exponent < -1	upper value < -1.0004077237

Table 1: Strict parameter margins certified by rational intervals.

By 4.1, choose a fixed K_b large enough to retain a uniform positive proportion of every L-rough source interval. This choice is made before L.

For each N, let $S_N \subseteq [N/2, N]$ consist of the L-rough integers satisfying the centered (A_b, K_b) condition. For fixed L and all sufficiently large N,

$$|S_N| \geq \frac{1}{4} N \rho_L.$$

Delete every $b \in S_N$ having a witness $d \leq N$, $d \neq b$, such that

$$b + d \mid 2bd, \qquad \Omega(d) \leq \Omega(b),$$

and call the surviving set B_N. As observed after 4.2, B_N is 2-admissible.

Because $A_b \log 4 < 3/2$, 4.2 permits L to be chosen, after K_b, so large that the first loss is at most $|S_N|/4$. The second residual range will be controlled when N is chosen after all constraints on L have been imposed. For all sufficiently large such N, we then have

$$|B_N| \geq \frac{1}{2} |S_N| \geq \frac{1}{8} N \rho_L. \tag{23}$$

Now put

$$K_o = \lceil c \log \log L \rceil$$

and let O_N be the odd $a \leq 2N$ satisfying the centered (A_o, K_o) condition. By 3.1 and (22),

$$N - |O_N| \ll N z_o^{-K_o} \ll N (\log L)^{-c \log z_o} = o_L(N \rho_L).$$

Enlarging the still-to-be-fixed cutoff L, we may arrange

$$N - |O_N| \leq \frac{1}{32} N \rho_L \tag{24}$$

for all sufficiently large N.

Let E_N be the number of $(a, b) \in O_N \times S_N$ with

$$a + 2b \mid ab.$$

The last inequality in (22) and 5.2 give

$$E_N \ll q_b^{K_b} q_o^{K_o} \left\{ N (\log L)^{-2} + o_L(N) \right\}.$$

Since K_b is fixed and

$$q_o^{K_o} \ll (\log L)^{c \log q_o}, \qquad c \log q_o < 1,$$

the first term is $o_L(N\rho_L)$. Enlarge L once more to control it. After L, and hence K_o, is fixed, take N large enough simultaneously for the rough source asymptotic, the second source residual range, the odd-host bound, and the normalized mixed o_L-term. Thus

$$E_N \leq \frac{1}{32} N\rho_L. \tag{25}$$

The choices have occurred in the legal order

$$\text{fixed real parameters} \longrightarrow K_b \longrightarrow L \longrightarrow K_o \longrightarrow N.$$

Let

$$\Gamma_N = \{a \in O_N : \text{some } b \in B_N \text{ satisfies } a + 2b \mid ab\}.$$

Since $B_N \subseteq S_N$, $|\Gamma_N| \leq E_N$. Define

$$A_N = (O_N \setminus \Gamma_N) \cup \{2b : b \in B_N\}.$$

Two odd members cannot conflict because their sum is even and their product is odd. Two even members $2b, 2d$ conflict exactly when $b + d \mid 2bd$, excluded by the 2-admissibility of B_N. Every mixed conflict was deleted in the definition of Γ_N. Hence A_N is 1-admissible.

Finally, (23), (24), and (25) give

$$\begin{aligned}
|A_N| &= |O_N| - |\Gamma_N| + |B_N| \\
&\geq N - \frac{1}{32} N\rho_L - \frac{1}{32} N\rho_L + \frac{1}{8} N\rho_L \\
&= N + \frac{1}{16} N\rho_L.
\end{aligned}$$

The selected L is now an absolute constant, so $\eta = \rho_L/16 > 0$ proves 1.2. □

Proof of 1.1. As noted after the theorem statement, apply 1.2 with $N = \lfloor X/2 \rfloor$. For all sufficiently large X,

$$f_1(X) \geq (1 + \eta) \lfloor X/2 \rfloor \geq \left(\frac{1}{2} + \frac{\eta}{4} \right) X.$$

□

8 Formal verification and reproducibility

The numerical supplement checks the exact mixed coordinates in finite boxes, the root count $3p - 2$, the complete small- and large-prime mixed Euler factors, the rational interval bounds in 1, and the symbolic cancellation leaving $(\log L)^{-2}$. These checks are not evidence for the analytic mean-value theorems; those enter only through the cited results in 2, whose hypotheses and cutoff uniformity are verified in the proofs above.

The accompanying Lean development formalizes the complete argument, including the analytic estimates. Rather than postulating the two cited mean-value theorems, it proves the special cases needed here: centered maximal prime-factor tails, explicit Mertens-type product bounds, the local three-linear-form weights, finite upper-sieve estimates with scheduled truncation errors, and the source and mixed dyadic summations. The transition and small-residual ranges are kept literal until their separate terminal estimates are proved.

The principal public declarations are

`Erdos327.Analytic.erdos327Conclusion_unconditional`,
`Erdos327.Analytic.erdos327SecondConclusion_unconditional`,
`Erdos327.Analytic.erdos327FullConclusion_unconditional`.

Their target propositions are the literal asymptotic statements in `Erdos327/Defs.lean`. The source conclusion is assembled in `SourceFinalSummation.lean`; the mixed bulk, error, terminal, and boundary budgets are combined in `MixedFinalSummation.lean`; and the common cutoff and final four-estimate construction are assembled in `Unconditional.lean`.

With Lean 4.33.0-rc1 and the pinned Mathlib revision, the root build checks all declarations. A project-wide scan contains no `sorry`, `admit`, project-local axiom, opaque theorem interface, or unsafe declaration. Lean's `#print axioms` reports only `propext`, `Classical.choice`, and `Quot.sound` for each of the three declarations above.

A literature sweep completed on 29 July 2026 located no prior paper or public proof claim resolving the first question. This is a dated search report, not a categorical priority claim. The centered source estimate is a refinement of Sawin's method, and the multivariable upper bound is due to de la Bretèche and Tenenbaum. A separate search located no prior public Lean formalization of Problem 327 or Sawin's result; existing public developments provide only more general prerequisites or incomplete nearby statements.

Errata relative to version 1

Two steps in the first released version of this manuscript were stated without adequate justification. Both are corrected above, and neither changes any theorem, exponent, or certified margin.

1. The assertion that every relevant dyadic block satisfies $X \gg_\delta L$ was used in 4.2 and 5.2 without proof. It is now proved: for the source, after (11); for the two-endpoint estimate, as (13). Both proofs run through the exclusion of $u = 1$, and in the two-endpoint case that exclusion uses $k = 1$ in an essential way, through the fact that x itself, and not merely $x/(x, k)$, must divide the greatest common divisor. The corresponding claim is false for a general odd multiplier, and the companion manuscript on the multipliers $k \geq 2$ has to change the dilation to repair it.

2. The identity $\Omega_L(a, x) = \Omega_L(t, x) + \Omega_{\geq L}(w) + \Omega(x)$ is only an inequality $\geq \cdots - 1$, since w may carry one prime factor exceeding x. This is now (14). The cost is the single absolute factor q_o in (15), absorbed by the implied constant; no exponent and no entry of 1 changes.

The Lean development is unaffected: it proves the analytic estimates in the special cases it needs rather than transcribing these two lines, and it was not rebuilt for this revision.

Computational and AI-use disclosure

AI systems were used for data analysis, reference search, adversarial proof audits, independent reconstruction of intermediate estimates, formalization assistance, and proofreading. The two-endpoint strategy, parameter choice, mathematical argument, and manuscript are the author's work. The author has checked the displayed calculations against the cited primary sources and accepts responsibility for all claims and errors.

References

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